

SURDS AND INDICES

Directions for questions 1 to 3: Select the correct alternative from the given choices.

$(12/17)^{15} \div (17/12)^{14}$ is equal to ____.

☒ a) $(12/17)^{29}$ (Your answer - ✓)

☐ b) $12/17$

☐ c) $17/12$

☐ d) $(17/12)^{29}$

Solution:

$$\left(\frac{12}{17}\right)^{15} \div \left(\frac{17}{12}\right)^{14} = \frac{12^{15}}{17^{15}} \frac{12^{14}}{17^{14}} = \frac{12^{29}}{17^{29}}$$

Choice (A)

Directions for questions 1 to 3: Select the correct alternative from the given choices.

$(7 \times 4 \times 3)^{5n}$ is equal to ____.

☒ a) $7^{5n} + 4^{5n} + 3^{5n}$ (Your answer - ✗)

☐ b) $(84^5)^n$ (Correct Answer - ✓)

☐ c) $(84)^{5^n}$

☐ d) $(84)^{n^5}$

Solution:

$$(7 \times 4 \times 3)^{5n} = (84)^{5n} = (84^5)^n \quad \text{Choice (B)}$$

Directions for questions 1 to 3: Select the correct alternative from the given choices.

Simplify $\frac{\sqrt{17}-\sqrt{11}}{\sqrt{17}+\sqrt{11}}$.

☐ a) $\frac{3}{14-\sqrt{187}}$

☐ b) $\frac{6}{\sqrt{17}-\sqrt{11}}$

☒ c) $\frac{6}{\sqrt{17}+\sqrt{11}}$ (Your answer - ✕)

☐ d) $\frac{14-\sqrt{187}}{3}$

Solution:

$$\frac{\sqrt{17}-\sqrt{11}}{\sqrt{17}+\sqrt{11}} \times \frac{(\sqrt{17}-\sqrt{11})^2}{(\sqrt{17}-\sqrt{11})^2} = \frac{(\sqrt{17}-\sqrt{11})^2}{17-11} = \frac{28-2\sqrt{187}}{6} = \frac{14-\sqrt{187}}{3}$$

Choice (D)

Directions for question 4: Type in your answer in the input box provided in the question.

a, b, c are positive numbers greater than 1 and $\sqrt{y} = a$, $\sqrt[3]{z} = b$, $\sqrt[4]{x} = c$. The value of

$\log_a x \times \log_b y \times \log_c z$ is

Your Answer: Unattempted

Correct Answer: 24 ✓

Solution:

$$\begin{aligned} & \log_{y^{1/2}} x \cdot \log_{z^{1/3}} y \cdot \log_{x^{1/4}} z \\ &= (2 \log_y x) (3 \log_z y) (4 \log_x z) \\ &= 24 \log_y x \log_z y \log_x z \\ &= 24 \end{aligned}$$

Ans : (24)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

If $\frac{8^n \times 2^3 \times (16)^{-1}}{2^n \times 4^2} = \frac{1}{4}$, the value of n is _____.

☐ a) 3

☒ b) $\frac{3}{2}$ (Your answer - ✓)

☐ c) 1

☐ d) $\frac{2}{3}$

Solution:

$$\frac{8^n 2^3 16^{-1}}{2^n 4^2} = \frac{2^{3n+3}}{2^{n+4+2}} = \frac{2^{3n+3}}{2^{n+6}} = 2^{2n-3} = 2^{-2} \text{ (Given)}$$

$$\therefore 2n - 3 = -2 \Rightarrow n = \frac{1}{2}$$

Choice (B)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

Let $\sqrt[3]{x} = a$. Find the value of $\log_a \sqrt[3]{x \sqrt[3]{x \sqrt[3]{x \sqrt[3]{x}}}}$

- ☐ a) $\frac{40}{27}$
- ☐ b) $\frac{20}{27}$
- ☐ c) $\frac{40}{9}$
- ☐ d) $\frac{40}{81}$

Solution:

Let the given expression be $\sqrt[3]{A}$, where

$$A = x \sqrt[3]{B}$$

$$B = x \sqrt[3]{C}$$

$$C = x \sqrt[3]{x}$$

$$\therefore C = x^{4/3}, B = x^{13/9}, A = x^{40/27} \text{ and } \sqrt[3]{A} = x^{40/81}$$

$$\therefore \text{The required value is } \log_{x^{1/3}} x^{40/81} = \frac{40}{81} (3) = \frac{40}{27}$$

Choice (A)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

What is the positive square root of $75 + 4\sqrt{176}$?

☐ a) $8 + \sqrt{11}$ (Correct Answer - ✓)

☐ b) $8 - \sqrt{11}$

☐ c) $\sqrt{37} + \sqrt{38}$

☐ d) $2\sqrt{5} + \sqrt{55}$

Solution:

$$75 + 4\sqrt{176} = 75 + 2\sqrt{704} = 64 + 11 + 2\sqrt{64(11)}$$

The square root is $8 + \sqrt{11}$.

Choice (A)

Directions for question 8: Type in your answer in the input box provided in the question.

If $\log_3(\log_4(\log_5 a)) = 0$ then $a =$

Solution:

$$\log_3(\log_4(\log_5 a)) = 0 = \log_3 1$$

$$\Rightarrow \log_4(\log_5 a) = 1 = \log_4 4$$

$$\Rightarrow \log_5 a = 4$$

$$\Rightarrow 5^4 = a$$

$$\Rightarrow a = 625$$

Ans : (625)

Directions for question 9: Select the correct alternative from the given choices.

$$\sqrt{176} + \sqrt{44} - \sqrt{99} = \underline{\hspace{2cm}}.$$

☐ a) $\sqrt{44}$

☒ b) $\sqrt{99}$ (Correct Answer - ✓)

☐ c) $\sqrt{88}$

☐ d) $\sqrt{22}$

Solution:

$$\sqrt{176} + \sqrt{44} - \sqrt{99} = 4\sqrt{11} + 2\sqrt{11} - 3\sqrt{11} = 3\sqrt{11} = \sqrt{99}.$$

Choice (B)

Directions for question 10: Type in your answer in the input box provided in the question.

If $\log(\cot 5^\circ) \log(\cot 15^\circ) \log(\cot 25^\circ) \dots \log(\cot 85^\circ) = a$, the value of a is

Your Answer: Unattempted

Correct Answer: 0 ✓

Solution:

$$\begin{aligned} \log(\cot 5^\circ) \log(\cot 15^\circ) \log(\cot 25^\circ) \dots \log(\cot 45^\circ) \dots \log(\cot 85^\circ) &= a \\ \Rightarrow \log(\cot 5^\circ) \dots \log(1) \dots \log(\cot 85^\circ) &= a \\ &\quad (\because \cot 45^\circ = 1) \\ \Rightarrow a &= 0 \quad (\because \log 1 = 0) \end{aligned}$$

Ans : (0)

Directions for question 11: type in your answer in the input box provided below the question.

Find the value of $\sqrt{42 + \sqrt{42 + \sqrt{42 + \dots \infty}}}$

Your Answer: Unattempted

Solution:

$$\text{Let } \sqrt{42 + \sqrt{42 + \sqrt{42 + \dots}}} = x.$$

$$\sqrt{42 + x} = x$$

$$\therefore x^2 - 42 = x$$

$$\Rightarrow x^2 - x - 42 = 0 \Rightarrow (x - 7)(x + 6) = 0 \Rightarrow x = 7$$

$$(\because x > 0)$$

Ans : (7)

Directions for question 12: Select the correct alternative from the given choices.

$$\left(\frac{x^{a+b}}{x^c}\right)^2 \times \left(\frac{x^{b+c}}{x^a}\right)^2 \times \left(\frac{x^{c+a}}{x^b}\right)^2 = \underline{\hspace{2cm}}.$$

☐ a) 0

☐ b) 1

☒ c) $x^{2(a+b+c)}$ (Your answer - ✓)

☐ d) $x^3(a^2 + b^2 + c^2) - ab - bc - ca$

Solution:

$$\begin{aligned} & \left(\frac{x^{a+b}}{x^c} \right)^2 \left(\frac{x^{b+c}}{x^a} \right)^2 \left(\frac{x^{c+a}}{x^b} \right)^2 \\ &= \left[\frac{x^{2(a+b+c)}}{x^{a+b+c}} \right]^2 = x^{2(a+b+c)} \end{aligned} \quad \text{Choice (C)}$$

Directions for question 13: Type in your answer in the input box provided in the question.

The value of y that satisfies the equation

$$\frac{\log 343}{\log 49}$$

$$= \log_9 y + 1 \text{ is}$$

Your Answer: Unattempted

Correct Answer: 3 

Solution:

$$\begin{aligned}\frac{\log 343}{\log 49} &= \log_e y + 1 \\ \Rightarrow \frac{3 \log 7}{2 \log 7} &= \log_e y + 1 \\ \Rightarrow \log_e y &= \frac{3}{2} - 1 = \frac{1}{2} \\ \Rightarrow y &= 9^{1/2} = 3\end{aligned}$$

Ans : (3)

Directions for questions 14 and 15: Select the correct alternative from the given choices.

$[16^{-1/2} - 4(8 - 6)^{-1}]^{-2}$ is equal to ____.

☐ a) $\frac{49}{64}$

☐ b) $\frac{1}{144}$

☒ c) $\frac{16}{49}$ (Correct Answer - ✓)

☐ d) $\frac{64}{49}$

Solution:

$$\begin{aligned}\left[16^{-\frac{1}{2}} - 4(8 - 6)^{-1}\right]^{-2} &= \left(\frac{1}{4} - \frac{4}{2}\right)^{-2} \\ &= \left(\frac{-7}{4}\right)^{-2} = \left[\left(\frac{-7}{4}\right)^{2-1}\right]^{-2} = \frac{16}{49}\end{aligned}$$

Choice (C)

Directions for questions 14 and 15: Select the correct alternative from the given choices.

Which of the following is the greatest?

$\sqrt{15} + \sqrt{10}$, $\sqrt{11} + \sqrt{14}$, $\sqrt{20} + \sqrt{5}$, $\sqrt{8} + \sqrt{17}$

☒ a) $\sqrt{8} + \sqrt{17}$ (Your answer - ✖)

☐ b) $\sqrt{20} + \sqrt{5}$

☐ c) $\sqrt{11} + \sqrt{14}$

☐ d) $\sqrt{15} + \sqrt{10}$

Solution:

Let $a = \sqrt{15} + \sqrt{10}$, $b = \sqrt{11} + \sqrt{14}$, $c = \sqrt{20} + \sqrt{5}$, $d = \sqrt{8} + \sqrt{17}$

We rearrange these as follows.

$$c = \sqrt{5} + \sqrt{20}$$

$$d = \sqrt{8} + \sqrt{17}$$

$$a = \sqrt{10} + \sqrt{15}$$

$$b = \sqrt{11} + \sqrt{14}$$

The sum of the two numbers in each of the 4 expressions is constant. As the smaller number increases, the expression would increase, i.e., $c < d < a < b$. The increase from $\sqrt{5}$ to $\sqrt{8}$ is more than the decrease from $\sqrt{20}$ to $\sqrt{17}$ and so on.

$\therefore$ b is the greatest.

Choice (C)